



SOFTWARE REQUIREMENTS SPECIFICATION

Point of Sale (POS) System

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Prepared For	Coders Lab
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1 Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the functional and non-functional requirements of the Cafe Management System. The system is intended to support the daily operations of a cafe by providing separate interfaces for Cashiers and the Cafe Owner. It will manage customer orders, tables, billing, menu items, and ingredient inventory in a centralized system.

This document provides the development team, QA/testerConventions owner with a shared and unambiguous understanding of what the system must do before design and implementation begin.

1.2 Document Conventions

- Requirements are labeled using the format FR-<Module>-<Number> for functional requirements and NFR-<Number> for non-functional requirements.
- Priority levels follow MoSCoW: Must Have, Should Have, Could Have, Won't Have (this release).
- The terms Cashier and Cafe Owner refer to the two primary user roles of the system.

1.3 Intended Audience and Reading Suggestions

- Developers — read Sections 3, 4, and 5 in full.
- QA / Testers — read Section 3 (for test-case derivation) and Section 5 (for NFR test plans).
- Project Managers / Stakeholders — read Sections 1 and 2 for scope and overview.
- UI/UX Designers — read Sections 2, 3, and 4.1.

1.4 Project Scope

The Cafe System will help cashiers manage orders, tables, bills, and payments, while the Cafe Owner can manage menu items and ingredient inventory. The system aims to reduce manual work, billing errors, and table conflicts.

In scope for v1.0:

- Create dine-in customer orders and add menu items to receipts.
- Add more items to an existing customer's receipt before payment.
- Increase the quantity of an ordered item.
- Assign orders to available tables and track table status.
- Release a table after the bill is paid.
- Add, edit, and remove menu items.
- Manage ingredients, including name, cost, stock level, and stock status.

- Generate bills based on ordered items and quantities.
- Apply valid discounts.
- Show the final payable amount.
- Complete an order after successful payment.
- Provide separate interfaces for Cashier and Cafe Owner.
- Allow the owner to manage menu and ingredient information.
- Allow the cashier to manage orders, tables, bills, and payments.

Out of scope for v1.0:

- Customer loyalty/rewards and mobile app.
- Food-delivery platform integration.
- Payroll and shift scheduling.
- Multi-branch management.

1.5 References

- UI/UX Wireframes — <https://pin.it/3Y1mSMaoJ>
- IEEE 830-1998 — Recommended Practice for Software Requirements Specifications

2 Overall Description

2.1 Product Perspective

The Cafe Management System will be a web-based application designed specifically for cafe operations.

The system will provide two role-based user interfaces:

Cashier Interface : designed for fast and simple handling of customer orders, table assignment, receipts, and bill processing.

Cafe Owner Interface : designed for managing menu items and managing cafe ingredients and stock information.

The system will replace or reduce manual processes used for recording orders, managing tables, calculating bills, and tracking cafe ingredients.

2.2 Product Features (Summary)

- Order Management — create orders, add items to receipts, and update item quantities.
- Table Management — assign orders to tables, track table status, and release tables after payment.
- Bill & Payment Management — generate bills, calculate discounts, and complete payments.
- Menu Management — add, edit, and remove menu items.
- Ingredient Inventory — track ingredients, costs, stock levels, and stock status.
- User Management — role-based access for Cashiers and Cafe Owners.
- Reporting — view basic sales and inventory information.

2.3 User Classes and Characteristics

User Class	Description	Importance
Cashier	Front-line staff who process customer orders, add items to receipts, assign tables, and handle bill payments. Requires a fast, simple, and easy-to-use interface for daily transactions.	High
Cafe Owner	Manages the cafe's menu and ingredient inventory, including menu items, ingredient costs, stock levels, and stock status. Needs a clear management and monitoring interface.	High

2.4 Operating Environment

- **Client application:** Web browser running on POS terminals, using a React.js frontend.
- **Server:** Hosted on Vercel.

- **Database:** Supabase PostgreSQL.
- **Peripherals:** Receipt printer, cash drawer, and card reader

2.5 Design and Implementation Constraints

- The system must maintain consistency between orders, table status, and payment status to prevent incorrect or duplicate transactions.
- An occupied table must not be assigned to another active order, and a table can only be released after the associated bill has been successfully paid.
- The system must maintain accurate bill calculations, including item quantities, VAT, discounts, and the final payable amount.
- Inventory information must remain consistent with the café's ingredient records, including stock quantity, cost, and stock status.
- The system must enforce role-based access so that Cashiers and Café Owners can access only the functions appropriate to their roles

2.6 Assumptions and Dependencies

- The cafe will have a reliable internet connection for normal system operation, with occasional connectivity interruptions expected.
- Supabase services will be available and accessible for storing and retrieving system data.
- Cashiers and the cafe owner will have valid login credentials and the required permissions for their respective interfaces.
- The cafe owner will provide accurate menu, ingredient, pricing, and inventory information.
- The receipt printer, cash drawer, and card reader will be available and compatible with the POS environment when required.

3 Cafe System (Functional Requirements)

Each feature below includes a short description, a MoSCoW priority, and a table of testable, numbered requirements. Use this structure for every module you add.

3.1 User Authentication & Roles

Allows Cashier and Owner to securely log in and restricts access to modules based on assigned role.

Priority: Must Have

Req ID	Requirement Description	Priority
FR-AUT H-01	The system shall allow login via username/password.	Must Have
FR-AUT H-02	The system shall lock an account for 15 minutes after 5 consecutive failed login attempts.	Must Have
FR-AUT H-03	The system shall restrict access to Ingredient Inventory, Menu Management, and Profit Calculation modules to the Admin/Owner role only.	Must Have
FR-AUT H-04	The system shall allow Cashier roles to access Order, Table, and Billing modules only.	Must Have
FR-AUT H-05	System shall auto logout an idle session after 20 minutes.	Should Have

3.2 Order Management

Allows creating new orders, adding items to a receipt, and adding new items to the same customer's open order.

Priority: Must Have

Req ID	Requirement Description	Priority
FR-ORD-01	The system shall allow creating a new order (Dine-in / Takeaway / Delivery).	Must Have
FR-ORD-02	The system shall allow adding menu items to an open order via item catalog.	Must Have
FR-ORD-03	The system shall allow adding new items to an existing receipt of the same customer/table without creating a duplicate order.	Must Have
FR-ORD-04	The system shall allow item quantity edit and item removal before payment.	Must Have
FR-ORD-05	System shall support item-level notes (e.g., "less sugar", "more ice").	Should Have

3.3 Table Management

Allows assigning orders to empty tables and automatically releasing a table once its bill is paid.

Priority: **Must Have**

Req ID	Requirement Description	Priority
FR-TBL-01	System shall display table status (Empty / Occupied / Reserved / Needs Cleaning).	Must Have
FR-TBL-02	System shall allow assigning an open order to an empty table.	Must Have
FR-TBL-03	System shall automatically release a table (set to Empty) once its bill is fully paid.	Must Have
FR-TBL-04	System shall allow merging two tables into one order for large groups.	Should Have
FR-TBL-05	System shall allow transferring an order from one table to another.	Could Have

3.4 Billing & Payment Management

Calculates the total bill, applies VAT/discounts, and processes payment completion across multiple methods.

Priority: Must Have

Req ID	Requirement Description	Priority
FR-BILL-01	The system shall auto-calculate subtotal, VAT (configurable %), service charge (if any), and grand total.	Must Have
FR-BILL-02	The system shall apply any discount as given by the owner to the entire bill.	Should Have
FR-BILL-03	The system shall accept cash payment and calculate change due.	Must Have
FR-BILL-04	The system shall support payments via mobile banking systems and cash.	Must Have
FR-BILL-05	The system shall print or generate a digital receipt within 3 seconds of payment confirmation.	Must Have
FR-BILL-06	System shall allow cancelling an order before payment is finalized (with reason logged, since Admin=Cashier this is for accountability).	Should Have

3.5 Inventory Management

Tracks the ingredient catalog, cost, and stock levels, and flags stock status as Low/Moderate/High.

Priority: Must Have

Req ID	Requirement Description	Priority
FR-INV-01	The system shall maintain an ingredient catalog with name, unit (kg/L/pcs), cost per unit, and current stock.	Must Have
FR-INV-02	The system shall auto-classify stock status as Low / Moderate / High based on configurable thresholds.	Must Have
FR-INV-03	The system shall link each menu item to a "recipe" (Bill of Materials)—i.e., which ingredients and how much of each go into one serving.	Must Have

FR-INV-04	The system shall automatically decrement ingredient stock based on the recipe whenever a menu item is sold.	Must Have
FR-INV-05	The system shall alert when an ingredient's stock falls below its reorder threshold.	Should Have
FR-INV-06	The system shall allow manual stock adjustment.	Must Have

3.6 Menu Management

Allows adding, editing, or removing menu items and linking each item to its ingredient recipe.

Priority: Must Have

Req ID	Requirement Description	Priority
FR-MEN U-01	The system shall allow adding, editing, and deactivating menu items.	Must Have
FR-MEN U-02	The system shall allow grouping items into categories (Coffee, Tea, Snacks, Desserts, etc.).	Must Have
FR-MEN U-03	System shall allow linking a menu item to its ingredient recipe (see FR-INV-03) for automatic cost calculation.	Must Have
FR-MEN U-04	The system shall support marking an item as "Out of Stock" (auto or manual), hiding it from ordering.	Should Have
FR-MEN U-05	The system shall support size/variant options (e.g., Small/Medium/Large) with different pricing.	Could Have
FR-MEN U-06	The system shall provide a read-only digital menu board displaying item names, prices, and photos, with automatic updates when the menu is edited.	Could Have

3.7 Profit Calculation

Calculates cost of goods sold (COGS) and profit margin per item and across the business.

Priority: Must Have

Req ID	Requirement Description	Priority
FR-PRO FIT-01	The system shall calculate Cost of Goods Sold (COGS) per item using its linked recipe cost (from FR-INV/FR-MENU).	Must Have
FR-PRO FIT-02	System shall calculate per-item and per-order profit margin (Sale Price – COGS).	Must Have

FR-PRO FIT-03	The system shall generate a daily/weekly/monthly profit summary (Revenue – COGS – optional fixed costs).	Should Have
FR-PRO FIT-04	The system shall show top-selling and most profitable items.	Could Have

3.8 Reporting

Provides sales summaries and performance visibility to support profit tracking.

Priority: **Should Have**

Req ID	Requirement Description	Priority
FR-REP -01	The system shall generate a daily sales summary report (total sales, transactions, average basket size).	Must Have
FR-REP -02	The system shall allow filtering reports by date range and category.	Should Have
FR-REP -03	The system shall allow exporting reports as CSV/PDF.	Should Have

4 External Interface Requirements

4.1 User Interfaces

- Sales/Order screen shall present large, touch-friendly buttons suitable for a 10–15 inch touchscreen terminal (Cashier station).
- A separate, simplified dashboard view (Inventory, Menu, Reports, Profit) shall be accessible to the Owner role, optimized for both touchscreen and desktop/laptop use.
- Color and iconography shall follow
- Menu items shall be organized by category tabs (Coffee, Tea, Snacks, Desserts, etc.) with item photos for quick visual selection.

4.2 Hardware Interfaces

- Receipt printer: [model], connected via USB / network, ESC/POS command set — for customer receipts.
- Kitchen Order Ticket (KOT) printer: [model, optional], connected via network — prints order tickets to the kitchen/counter station so items can be prepared without the barista needing to see the main screen.
- Cash drawer: triggered automatically by the receipt printer on cash sale completion.
- Card/mobile-payment (bKash/Nagad) reader or QR display: [model/provider], connected via [USB / Bluetooth / network].

4.3 Software Interfaces

- Frontend — React (SPA/PWA) for Cashier and Owner UIs.
- Backend — Node.js + Express.js REST API for Order, Table, Billing, Inventory, Menu, and Profit/Report modules.
- Database — Supabase (PostgreSQL), accessed via Supabase JS client.
- Authentication — Supabase Auth for Cashier/Owner login and role management.
- Realtime sync — Supabase Realtime for live table-status and stock updates across terminals.
- File storage — Supabase Storage for menu item images.
- Payment Gateway API — [bKash/Nagad/SSLCommerz/Card, name & version] for authorizing payments.
- Accounting export — [optional] CSV/PDF export of daily sales totals.

4.4 Communication Interfaces

- Frontend–Backend and Backend–Supabase communication over HTTPS (TLS 1.2+).
- Real-time updates across terminals via Supabase Realtime (polling as fallback).

5 Non-Functional Requirements

5.1 Performance

Req ID	Requirement Description	Priority
NFR-01	A checkout transaction (from finalizing the bill in Bill Payment Management to receipt generation) shall complete in under 3 seconds under normal load.	Must Have
NFR-02	The system shall support at least 10 concurrent terminal sessions per store without degradation.	Should Have
NFR-03	The Order Management screen shall load the current menu catalog and item availability within 1 second.	Should Have

5.2 Security

Req ID	Requirement Description	Priority
NFR-04	Where card payments are processed via a hosted/tokenized gateway (e.g., bKash, Nagad, or a card processor's hosted checkout) such that raw card data never reaches the application backend, the system shall meet PCI-DSS SAQ-A scope requirements; no card numbers shall be stored locally.	Must Have
NFR-05	Authentication credentials shall be managed via Supabase Auth; the system shall never store plaintext PIN/password values in the application database.	Must Have
NFR-06	All API traffic — between the client and the Express backend, and between the Express backend and Supabase — shall be encrypted using TLS 1.2 or higher.	Must Have
NFR-07	The system shall enforce role-based access control, restricting menu edits, discount application, and stock adjustments to the Owner role.	Must Have

5.3 Usability

Req ID	Requirement Description	Priority
NFR-08	A new cashier shall be able to complete a standard sale after no more than 30 minutes of training.	Should Have
NFR-09	The system shall display clear, human-readable error messages (no raw error codes) to end users.	Should Have

5.4 Reliability & Availability

Req ID	Requirement Description	Priority
NFR-10	The system shall target 99.5% uptime during store operating hours, dependent on hosting and Supabase's own SLA.	Could Have
NFR-11	The system shall continue to process cash sales and table/order updates in offline mode, and shall sync automatically (including ingredient stock deductions) once reconnected.	Should Have

5.5 Scalability & Maintainability

Req ID	Requirement Description	Priority
NFR-12	The system architecture shall support adding new store locations without code changes to core modules.	Could Have
NFR-13	The codebase shall maintain at least 50–60% automated test coverage on core order, table, and payment modules for the MVP, with a target of increasing coverage in later versions.	Should Have

6 Other Requirements

- **Legal/Compliance:** VAT rate shall be a configurable value (not hardcoded), set according to the cafe's classification (e.g., AC vs. non-AC) per current NBR rules—confirm the applicable rate with an accountant, as it has changed more than once in recent years
- **Fiscal Device Readiness:** The system shall be designed to support future integration with NBR-approved EFD/SDC infrastructure, should the business become subject to mandatory fiscal-device requirements as the rollout expands beyond highway restaurants.
- **Data Retention:** Transaction records shall be retained for a minimum of 5 years, per NBR record-keeping requirements for VAT purposes.
- **Localization:** The system shall support BDT currency and Bengali/English language, with the ability to add locales in future releases.

Appendix A: Glossary

Term	Definition
POS	Point of Sale — the location/system where a retail transaction is completed.
SKU	Stock Keeping Unit — a unique identifier for each distinct product.
PCI-DSS	Payment Card Industry Data Security Standard.
VAT	Value Added Tax
MoSCoW	Prioritization method: Must, Should, Could, Won't have.

Appendix B: Analysis Models

ERD: <https://app.eraser.io/workspace/kBg5X8ElJICAJEbVabnJ?origin=share>

User flow: <https://app.eraser.io/workspace/dOnqzWy1DBfMZSwhm3Yo?origin=share>

Appendix C: Open Issues

Issue ID	Description	Status
ISS-01	Which payment gateway will be supported by the POS in the store (e.g., bKash, Nagad, Rocket)?	Open
ISS-02	What VAT rate will be imposed on orders?	Open
ISS-03	Can the admin change menu item prices while the cafe is operating?	Open
ISS-04	Should ingredient stock automatically decrease when an order is completed/paid?	Open
ISS-05	Should menu items automatically become unavailable when required ingredients are out of stock?	Open